

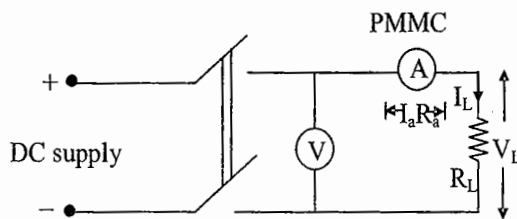
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Measurement of Power & Energy

- DC power $\rightarrow$ (Only for R-Load) $\Rightarrow P = VI$
 - Voltmeter – Ammeter method
 - Ammeter – voltmeter method
- AC power $\rightarrow R, L$ & C loads $\Rightarrow P = VI \cos \phi$
Measured by Dynamometer type wattmeter

Measurements of DC power

1. V – A method:



Conclusions

- $P_m > P_t$ (Measured power greater than true power)
- Error is due to ammeter only
- Suitable for measurement of high resistance and low load current

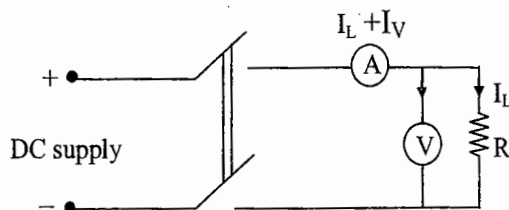
$$P_m = \text{voltmeter reading} \times \text{Ammeter reading}$$

$$= (V_L + V_a) \times I_L$$

$$= V_L I_L + V_a I_L$$

$$P_m = P_{\text{true}} + I_L^2 R_L \rightarrow \text{Power loss}$$

2. A – V method:



$$(i) P_m > P_t$$

(ii) Error is due to voltmeter

(iii) Suitable for measurement of low resistance and high load currents

$$P_m = V_L \times (I_L + I_v)$$

$$P_m = V_L I_L + V_L I_v$$

$$P_m = P_t + V_L I_v$$

Note:

In both the methods ($P_m > P_t$)

Note:

In both the methods the error due to always “Load side” of instruments

- AC power can be measured by using wattmeter as discussed previous lesson

Measurement of poly phase power:

3 - ϕ

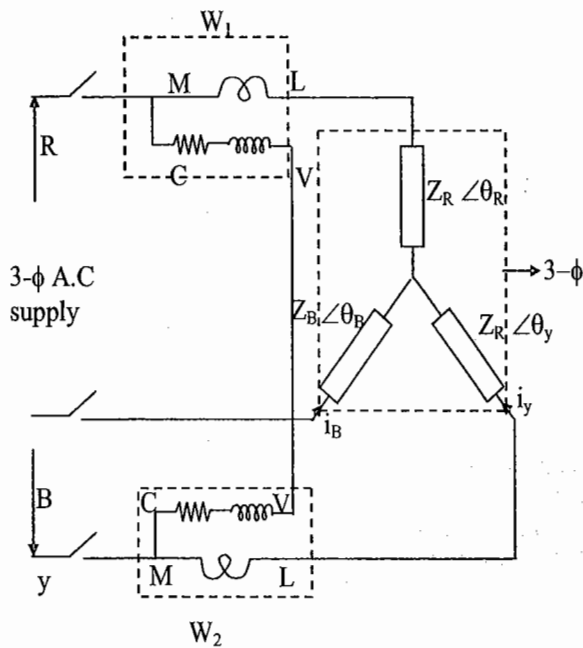
Balanced ($I_n = 0$) $Z_1 = Z_2 = Z_3$ & $\theta_1 = \theta_2 = \theta_3$

Unbalanced ($I_n \neq 0$)

3 - ϕ 4 wire			3 - ϕ 3 wire		
	Min	Max		Min	Max
Balanced	1	3	Balanced	2	2
unbalanced	2	3	Unbalanced	2	2

- For 3 ϕ 4- wire $\rightarrow$ neutral is present
- As per blondel's theorem, if neutral is not available for 'n' wire system then 'n-1' no. of wattmeter's are required for 'n' wire system.

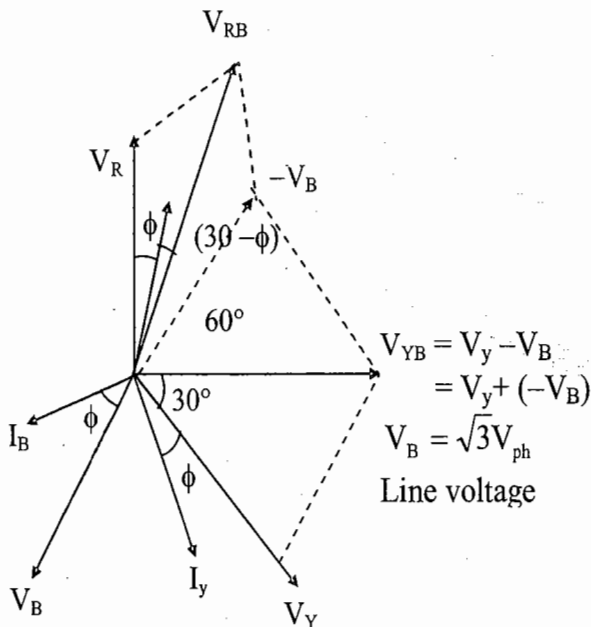
Consider 3- ϕ 'Y' connected lagging power factor balanced load with “R-Y-B” phase sequence ('B' phase made common)



3- ϕ Y- connected "lagging loads

$$W_1 = \vec{V}_{RB} I_R \cos \angle V_{RB} \times I_R$$

$$W_2 = \vec{V}_{YB} I_Y \cos \angle V_{YB} \times I_Y$$



$$W_1 = \vec{V}_{RB} \vec{I}_R \cos(30 - \phi)$$

$$W_2 = \vec{V}_{YB} \vec{I}_Y \cos(30 + \phi)$$

For balanced condition

$$V_{RB} = V_{YB} = V_L$$

$$I_R = I_Y = I_L$$

$$W_1 = V_L I_L \cos(30 - \phi)$$

$$W_2 = V_L I_L \cos(30 + \phi)$$

$$P = W_1 + W_2 = \sqrt{3} V_L I_L \cos \phi$$

In star connection

$$V_L = \sqrt{3} V_{ph}; I_L = I_{ph}$$

$$P = 3 V_{ph} I_{ph} \cos \phi \rightarrow \text{Total } 3\phi \text{ power}$$

$$W_1 - W_2 = V_L I_L \sin \phi$$

$$W_1 - W_2 = \sqrt{3} V_{ph} I_{ph} \sin \phi$$

$$V_{ph} I_{ph} \sin \phi = \frac{W_1 - W_2}{\sqrt{3}}$$

$$Q_1 - \text{phase} = \frac{W_1 - W_2}{\sqrt{3}}$$

Reactive power for 3 - ϕ

$$Q_T = 3 \left(\frac{W_1 - W_2}{\sqrt{3}} \right) = \sqrt{3} (W_1 - W_2)$$

Power factor angle

$$\frac{W_1 - W_2}{W_1 + W_2} = \frac{V_L I_L \sin \phi}{\sqrt{3} V_L I_L \cos \phi}$$

$$\frac{W_1 - W_2}{W_1 + W_2} = \frac{\tan \phi}{\sqrt{3}}$$

$$\phi = \tan^{-1} \left(\frac{\sqrt{3} (W_1 - W_2)}{(W_1 + W_2)} \right)$$

$$\text{p.f} = \cos \phi$$

Advantages of 2 - wattmeter method:

Note:

- (1) It can be used for balanced and unbalanced, whether neutral is available (or) not



Example 1:

The power flowing in a 3phase 3wire balanced load system is measured by two wattmeter method. The reading of wattmeter A is 7500W and of wattmeter B is -1500W ; then find what is the power factor of the system.

Sol: $P_1 = 7500W$

$$P_2 = -1500W$$

$$P = P_1 + P_2 = 7500 - 1500 = 6000W$$

$$P_1 - P_2 = 7500 + 1500 = 9000W$$

$$\phi = \tan^{-1} \left\{ \sqrt{3} \left[\frac{P_1 - P_2}{P_1 + P_2} \right] \right\}$$

$$= \tan^{-1} \sqrt{3} \left(\frac{9000}{6000} \right) = 69^\circ$$

$$p.f = \cos \phi = \cos 69^\circ = 0.358$$

Example 2:

A 3 ϕ , 500 V motor has a power factor of 0.4. Two wattmeter's connected to measure the input to be 30kW, Then the reading of each wattmeter.

(a) $W_1 = 34.85 \text{ kW}, W_2 = -4.85 \text{ kW}$

(b) $W_1 = 38.26 \text{ kW}, W_2 = -8.26 \text{ kW}$

(c) $W_1 = 69.68 \text{ kW}, W_2 = -39.681 \text{ kW}$

(d) $W_1 = 0 \text{ kW}, W_2 = 30 \text{ kW}$

Ans: (a)

Sol: $W_1 + W_2 = 30 \text{ kW} \text{ ----- (1)}$

$$\cos \phi = 0.4$$

$$\tan \phi = 2.291$$

$$\tan \phi = \frac{\sqrt{3}(W_1 - W_2)}{(W_1 + W_2)}$$

$$2.291 = \frac{\sqrt{3}(W_1 - W_2)}{30 \text{ kW}}$$

$$W_1 - W_2 = 39.68 \text{ kW} \text{ ----- (2)}$$

Solving equation (1) & (2)

$$W_1 = 34.84 \text{ kW}$$

$$W_2 = -4.85 \text{ kW}$$

Example 3:

The result of ratio of the readings of two watt meters connected to measure power in a balanced 3 - ϕ , 3 - wire load is 'r'. The load is known to be inductive with a lagging power factor. Find the power factor of the load?

(a) $\sqrt{0.2 + 3 \left(\frac{r-1}{r+1} \right)^2}$ (b) $\frac{1}{\sqrt{1 + 3 \left(\frac{r-1}{r+1} \right)^2}}$

(c) $\frac{r-1}{r+1}$ (d) $\frac{1}{\sqrt{2 + 3 \left(\frac{r-1}{r+1} \right)^2}}$

Ans: (b)

Sol: $\tan \phi = \sqrt{3} \left(\frac{W_1 - W_2}{W_1 + W_2} \right) \Rightarrow \sqrt{3} \left(\frac{\frac{W_1}{W_2} - 1}{\frac{W_1}{W_2} + 1} \right)$

$$\cos \phi = \frac{1}{\sec \phi}$$

$$= \frac{1}{\sqrt{1 + \tan^2 \phi}} \Rightarrow \frac{1}{\sqrt{1 + 3 \left(\frac{r-1}{r+1} \right)^2}} \because r = \frac{W_1}{W_2}$$



Readings of W_1, W_2 for different load p.f:

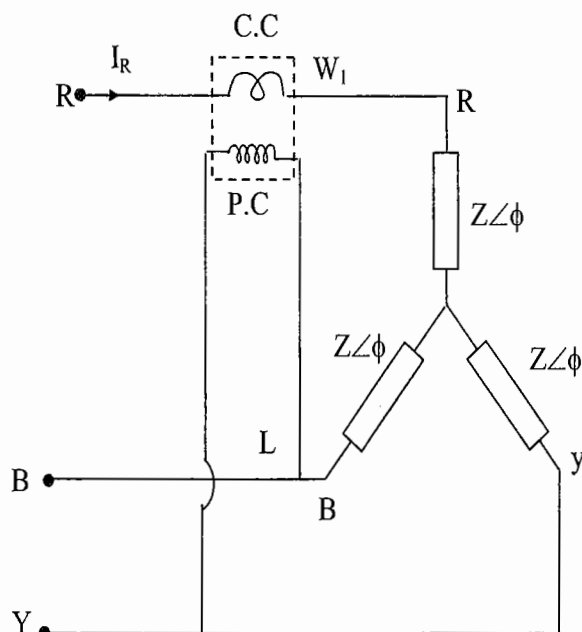
ϕ	$\cos\phi$	$W_1 = V_L I_L \cos(30 - \phi)$	$W_2 = V_L I_L \cos(30 + \phi)$	$W_1 + W_2$	Observations
0	1	$\frac{\sqrt{3}}{2} V_L I_L$	$\frac{\sqrt{3}}{2} V_L I_L$	$\sqrt{3} V_L I_L$	$W_1 = W_2$
30	0.866	$V_L I_L$	$\frac{V_L I_L}{2}$	$1.5 V_L I_L$	$W_2 = 2W_1$
60	0.5	$\frac{\sqrt{3}}{2} V_L I_L$	0	$\frac{\sqrt{3}}{2} V_L I_L$	$W_1 = 0$
90	0	$\frac{V_L I_L}{2}$	$-\frac{V_L I_L}{2}$	0	$W_1 = -ve$ $W_2 = +ve$

Note:

If one of the wattmeter shows negative then reverse either potential coils (or) current coil terminals and record the wattmeter reading with (-ve sign.)

Eg: Wattmeter reading is 20W after reversing current coil terminals then take the reading as 'W = -20W'.

Measurement of Reactive power:



$$W_1 = V_{YB} I_R \cos \angle V_{YB} - I_R$$

$$W_1 = V_{YB} I_R \cos(90 - \phi)$$

$$W = V_L I_L \sin \phi$$

$$W = \sqrt{3} V_{ph} I_{ph} \sin \phi$$

$$Q_T = \sqrt{3} (W) = 3 V_{ph} I_{ph} \sin \phi \rightarrow 3 - \phi \text{ reactive power}$$

NOTE:

- For measurement of reactive power the current coil is connected to one of the phases of a 3 - ϕ network
- Then the potential coil is to be connected between the remains two phases.

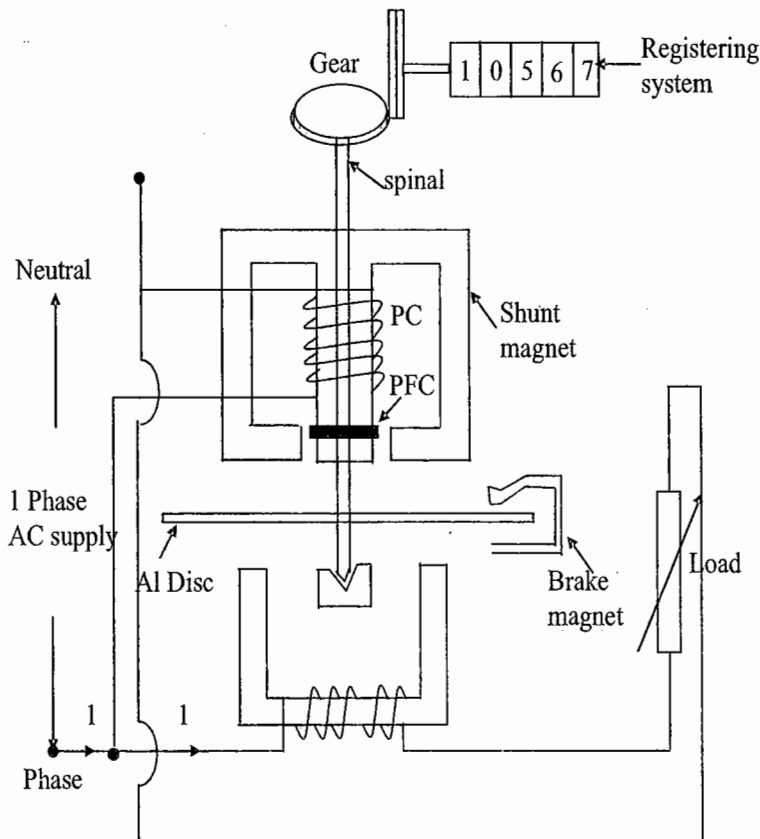
Measurement of Energy

The power delivered over a period of time is called energy

$$\text{Energy} = \int_0^t V I \cos \phi dt$$

1- ϕ energy meter induction type:

Energy meter is an integrating instrument which works on the principle of induction.



Energy meter consists of

- (i) Driving mechanism
- (ii) Rotating mechanism
- (iii) Breaking mechanism
- (iv) Recording (or) registering mechanism

Driving mechanism:

- (a) shunt magnet
- (b) series magnet

Shunt magnet:

Shunt magnet is "highly inductive and it has large no. of turns

- On shunt magnet potential coil is placed and it is directly connected to the supply
- P.C carries a current of I_p . It is proportional to supply voltage only and it is independent of load.

Series magnet:

It is wound with a coil called as "current coil" (CC) is connected in series with load and carries the current ' I_c ' proportional to load I .

Rotating mechanism:

A light weight aluminum disc is used and it is placed on spindle. The speed of aluminum disc is proportional to power consumed by the load

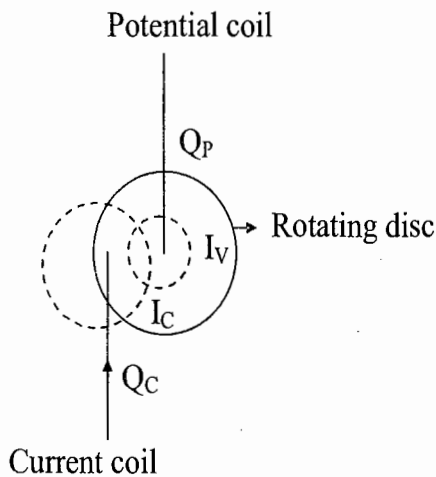
Operation:

Potential coil produces a flux (ϕ_p) and this flux is passing through the rotating disc. Which induces an eddy voltage (e_p) and an eddy current (I_{ep}). When current coil produces flux ϕ_c , which induces an eddy current I_{ec} in the disc. The interaction of (I_{ec}) with potential coil flux (ϕ_p) and

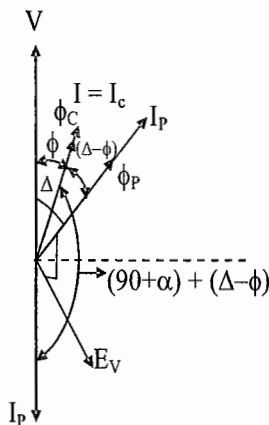
interaction of I_{ep} with current coil flux ' ϕ_c ' produces a torque in the rotating disc. This torque is proportional to the power consumed by the load.

Note:

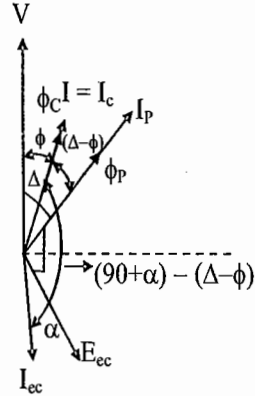
If this power consumption is measured over a time period (dt), then the no. of revolutions of rotating disc is proportional to the energy consumed by the load. But between two torques the angle is 90° . The resultant of T_{d1} and T_{d2} net driving torque (T_d) is produced, with this disc starts rotating



Phasor for potential coil emf:



Phasor for current coil emf:



$$T_{d1} = \phi_C I_{ep} \cos \angle \phi_c \text{ \& } I_{ep}$$

$$T_{d2} = \phi_P I_{ec} \cos \angle \phi_c \text{ \& } I_{ec}$$

$$T_{d1} = VI \cos(90 + \alpha) - (\Delta - \phi)$$

$$T_{d2} = VI \cos(90 + \alpha) + (\Delta - \phi)$$

$$T_d = T_{d1} - T_{d2}$$

$$= VI((\cos(90 + \alpha) - (\Delta - \phi)) - \cos((90 + \alpha) + (\Delta - \theta)))$$

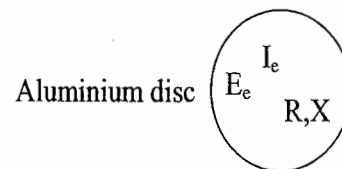
$$= VI(2 \sin(90 + \alpha) \sin(\Delta - \phi))$$

$$T_d = VI(2 \cos \alpha \sin(\Delta - \phi))$$

$$2 \cos \alpha \rightarrow \text{Al disc material}$$

$$T_d \propto VI \sin(\Delta - \phi)$$

- If all disc reactance $X = 0$, then driving torque will be maximum



$$T_d \propto VI \sin(\Delta - \phi) \rightarrow W_n(\text{measure value}) X = 0$$

$$T_d \propto VI \cos \phi \rightarrow W_T = (\text{True value}) \alpha = 0$$

$$\text{Error} = W_m - W_T = VI \sin(\Delta - \phi) - VI \cos \phi$$

$$\text{Error} = VI(\sin(\Delta - \phi) - \cos \phi)$$

$$\text{Error} = \frac{\sin(\Delta - \phi) - \cos \phi}{\cos \phi} \times 100$$

Example 4:

A correctly adjusted single phase 240V induction watt hour meter has a meter constant of 600 revolutions per kwh. Determine the speed of the disc, for a current of 10A at a power factor of 0.8 lagging. If the lag adjustment is altered so that phase angle between voltage coil flux and applied voltage is 86° calculate the error introduced at 0.5 pf lagging

- (a) 19.8rpm; -10% (b) 19.8rpm; +10%
(c) 19.2rpm; -11.8% (d) 19.2rpm; -12.3%

Ans: (d)

Sol: Energy consumed in one minute with rated current and 0.8pf lagging

$$240 \times 10 \times 0.8 \times \left(\frac{1}{60}\right) \times 10^{-3} = 0.032 \text{ kwh}$$

$$\therefore \text{Revolutions made in one minute} = 0.032 \times 600$$

$$= 19.2 \text{ rpm}$$

$$\therefore \text{speed of the DISC} = 19.2 \text{ rpm}$$

$$\text{Error introduced} = \frac{\sin(\Delta - \phi) - \cos \phi}{\cos \phi} \times 100$$

$$\text{We have } \Delta = 86^\circ$$

$$\text{p.f} = 0.5 \text{ lagging}$$

$$\therefore \text{error} = \frac{\sin(86^\circ - 60^\circ) - \cos 60^\circ}{\cos 60^\circ} \times 100$$

$$= -12.3\%$$

- Energy meter works on 1- ϕ I.M principle i.e. the angle between potential coil and cc flux is maintained 90° . Then two driving torque T_{d1} and T_{d2}

At steady state of a/d disc

$$\frac{k_3 \phi_m^2 N R}{Z} = T_d$$

$$N = T_d \frac{Z}{k_3 \phi_m^2 R}$$

$$N = V I \cos \phi$$

$$\int_0^t N dt \propto \int_0^t V I \cos \phi$$

↓ ↓
Mech. Electrical
Energy energy

No. of revolutions $\propto$ Electrical energy

Note:

- Because of braking torque, it runs at steady state

$$T_B \propto \phi_m \cdot d$$

$$N \propto \frac{T_B}{d \phi_m} \Rightarrow N \propto \frac{1}{d}$$

d = distance between spindle and permanent magnet

- To increase the speed permanent magnet is kept nearer to the spindle
- To decrease the speed magnet is kept far away from the spindle
- If angle (Δ) between "PC voltage (V) and PC coil flux (Q_p)" (or) I_p is 90° . Then energy meter records true power and error is zero.

Note:

In energy meter the nature of potential coil should be highly inductive and has large no. of turns, then error is zero.

Breaking torque:

$$E_e \propto \phi_m N$$

$$= k_1 \phi_m N$$

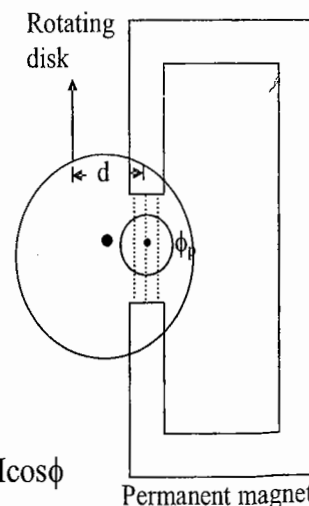
$$T_e \propto \phi_m I_e R$$

$$T_B = k_2 \phi_m I_e R$$

$$T_B = \frac{k_3 \phi_m^2 N R}{Z}$$

$$T_B \propto \phi_m^2$$

$$\therefore T_B \propto N, T_d \propto V I \cos \phi$$





N = speed

R = Radius of spindle and permanent magnet

Formulae in energy meter:

Energy = power × time

$$= VI \cos \phi \times t$$

$$= \frac{VI \cos \phi}{1000} \times \frac{t}{3600} \text{ kWhr}$$

(∵ t in sec, 1 kWhr = 1 unit At full load I = i,

at half load I = $\frac{i}{2}$)

Energy meter constant:

$$K = \frac{\text{No. of revolutions}}{\text{kWhr}}$$

$$K = \frac{N}{\text{pt}}$$

P = power kW

t = time in hr

N = speed of disc = revolutions

$$\text{Creeping error} = \frac{\text{Total no. of revolutions made}}{\text{Energy meter const } t(K)}$$

% Creeping error

$$= \frac{\text{No. of revolutions due to creeping}}{\text{Revolutions due to load}} \times 100$$

% Energy meter constant

$$= \frac{\text{Energy recorded}}{\text{True energy consumed}} \times 100$$

= +ve – fast

= -ve – slow

Example 5:

A single phase kwh meter makes 500 revolutions per kwh. It is found on testing as making 40 revolutions in 58.1 seconds at 5 kw full load. The percentage error in the meter is _____ (in %)

Ans: 0.86

Hint: Energy consumed (E.C)_t = kwh

$$= 5 \times \frac{58.1}{3600} = 0.0807 \text{ kwh}$$

$$(E.C)_m = \frac{\text{rev}}{M.C} = \frac{40}{500} = 0.08 \text{ kwh}$$

$$\% \text{ error} = \frac{0.08 - 0.0807}{0.0807} \times 100 = -0.86\%$$

Example 6:

Statement for Linked Answer Questions

A single phase energy meter is tested for half an hour run at a supply voltage of 230V and a load current of 10A at 0.8 lag power factor. The dial reading at start was 57.35 and at the end of the test was 58.25. The meter constant is 1200 revolutions per kwh. The meter made 890 revolutions in 30 minute run

(i) The error in meter registration is

- (a) 0.02 kwh more (b) 0.04 kwh less
(c) 0.02 kwh less (d) 0.2 kwh more

Ans: (c)

$$\text{Sol: (Energy consumption)}_t = \frac{V_L I_L \cos \phi}{1000} \times 0.5$$

$$= \frac{230 \times 10 \times 0.8}{1000} \times 0.5$$

$$= 0.92 \text{ kwh}$$

$$(E.C)_m = 58.25 - 57.35 = 0.90 \text{ kwh}$$

$$\therefore \text{error} = A_m - A_t = 0.90 - 0.92 = -0.02 \text{ kwh}$$

(ii) The percentage error in meter revolutions

- (a) 19.38% fast (b) 19.38% slow
(c) 17.21% fast (d) 1.28% slow

Ans: (b)

$$\text{Sol: (Rev)}_t = M.C \times E.C = 1200 \times 0.92 = 1104 \text{ rev}$$

$$(\text{Rev})_m = 890 \text{ rev}$$

$$\% \text{ error} = \frac{890 - 1104}{1104} \times 100 = -19.38\%$$

Creeping error:

When PC is energized and there is no current in current coil, then the disc revolves slowly and continuously. This effect is called creeping error.

Causes:

- Main reason is over static friction compensation
- Over voltages
- Excessive lubrications, vibrations, stray magnetic field of Instrument.
- To eliminate the creeping error, two holes are drilled on opposite sides of the disc diameter.
- A small Iron piece is attached to the edge of an disc, under creeping. Thin Iron piece is attracted by permanent magnet so disc stops creeping.

Example 7:

A 230V, 1 – ϕ watt hour meter has a constant load of 4A passing through it for 6 hours at unity p.f if the meter disc makes 2208 revolutions during this period. calculate the power factor of the load if the number of revolutions made by the meter are 1472 when operating at 230V and 5A for 4 hours.

- (a) 0.8 (b) 0.6
(c) 0.75 (d) 0.92

Ans:(a)

Sol: No. of revolutions = $kVI \cos \phi \times t \rightarrow (1)$

$$\therefore k = \frac{2208}{230 \times 4 \times 1 \times 6} = 400 \text{ rev/kwh}$$

$$\therefore \text{for 1472 revolutions } \cos \phi = ?$$

$$(1) \Rightarrow 1472 = (400 \times 230 \times 5 \times \cos \phi \times 4) \times 10^{-3}$$

$$\Rightarrow \cos \phi = 0.8$$

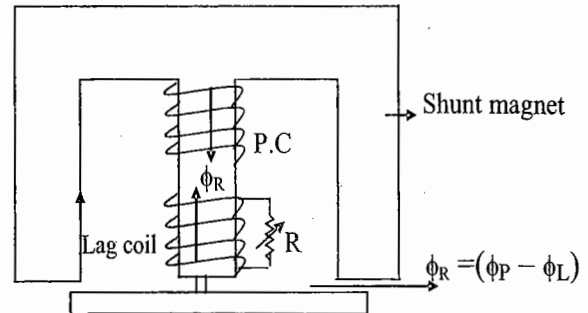
Phase lag/ Quadrature / Inductive /Lag coil

For the measurement of true energy by the energy meter potential coil current (I_P) lags behind the

voltage (V) by an angle 90° . But in practical it is Δ (You can see this Δ from phasor diagram of energy meter)

To make this angle 90° a Lag coil is connected below the potential coil which will produces flux ϕ_L .

As shown in below figure



By adjusting the lag coil (means by moving it up or down) we can make ϕ_R exactly in quadrature with the voltage. So that Iron losses in the core is small.

Example 8:

A single phase energy meter of the induction type is rated 230 V, 10A, 50 Hz and has a meter constant of 600 rev/Kwh when correctly adjusted. If quadrature adjustment is slightly disturbed so that the lag is 85° , calculate the percentage error at full load, 0.8 pf lag.

- (a) -5.5% (b) -6.9%
(c) -9.3% (d) -1.3%

Ans: (b)

Sol: $T_d \propto \sin \alpha$

Where α = angle between two fluxes.

$$\text{Given } \phi = \cos^{-1} (0.8) \Rightarrow 36.8$$

$\therefore$ The value of angle under measurement is

$$\alpha^1 = 85 - 36.8 \Rightarrow 48.2^\circ$$

$$\text{But true value is } \alpha = 90^\circ - 48.2$$

$$= 53.2$$

$$\therefore \text{error} = \frac{\sin \alpha^1 - \sin \alpha}{\sin \alpha} \times 100$$

$$= \frac{\sin 48.2^\circ - \sin 53.2^\circ}{\sin 53.2^\circ} \times 100$$

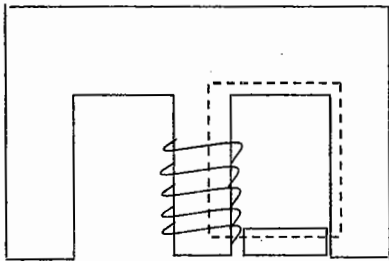
$$= -6.9\%$$

SHADING BANDS

Which are similar to the “lag coil” which are made of Cu. By adjusting the position of Shading Band ϕ_L is adjusted such that ϕ_R is kept at 90° w.r.t voltage.

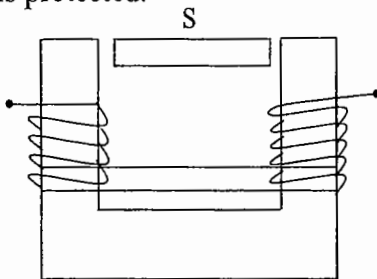
“OVER VOLTAGE” Compensation :-

Saturable shunt magnets are kept between side limb of the shunt magnet. Under normal air gap between saturable shunt magnet and the side limbs produces Maximum reluctance but under high voltage reluctance breaks so that flux is diverted hence meter is protected.



Over current protection

A saturable series magnet is kept between the side limbs of series magnet under heavy load currents flux diverts through this magnet hence meter is protected.



Temperature compensation:

The permanent magnet is made up of “Mu temperature” which is less affected by the surrounding temperature.

Testing of Energy Meter

Phantom/fictitious Loading :

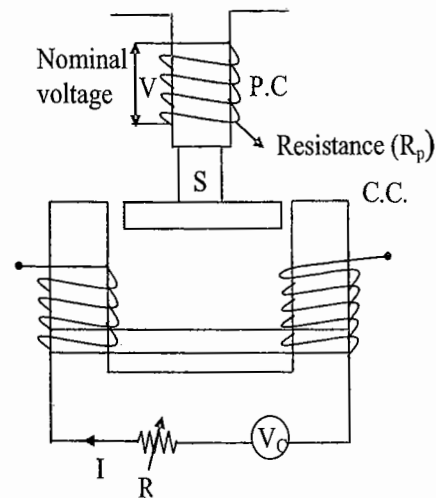
Phantom (or) fictitious loading is used to reduce the energy loss due to actual loading. So that the energy meter is tested at the desired load current and nominal supply voltage.

Procedure:

The potential coil of the energy meter is applied with normal supply voltage.

The current coil is supplied with a low voltage supply variable resistance is connected in series with it.

By varying resistance (R) we can make the current through current coil equal to the load current called fictitious loading.



Under Normal loading :

$$\text{Total power loss} = \frac{V^2}{R_p} + VI \quad \text{“I” is load current}$$

so due to normal loading energy loss is high.

Note: In energy meter potential coil (or) current coil terminals are reversed then disc rotates in the opposite direction.